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Code-to-code comparison between FLASH and HYDRA in gas-puff Z-pinch modeling

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ABSTRACT

The numerical modeling of gas-puff Z pinches involves the nonlinear coupling of a broad range of complex, multi-physics phenomena that makes such simulations challenging. The challenge is further compounded by nonlinear instabilities that can impact the dynamics of imploding gas-puff Z pinches, such as the magneto Rayleigh–Taylor instability (MRTI). If the growth rate and amplitude of the latter is comparable to the relevant timescales and properties of the imploding plasma, the MRTI can dramatically alter implosion dynamics, dictate pinch stability, and govern the plasma properties achievable in pulsed-power-driven laboratory experiments. National Laboratories and academic teams have developed numerical tools that can accurately model Z-pinch configurations and provide reliable design capabilities that can guide experimental choices and assist in interpreting experimental results. Most such tools, however, are not broadly available. Here, we present newly developed Z-pinch simulation capabilities of the publicly available FLASH code, applied in the study of MRTI growth and dynamical effects in gas-puff implosions. To verify the new implementations, we perform a comparison of FLASH gas-puff implosion simulations with previously published calculations with the HYDRA code from Lawrence Livermore National Laboratory, which have been validated with experimental data from the CESZAR pulsed-power driver at the University of California, San Diego. The experiments involved double- and triple-nozzle configurations, in an experimental attempt to stabilize the pinch to the MRTI. The code-to-code comparison shows similar results between the FLASH and HYDRA simulations, supporting the use of FLASH in the modeling of future gas-puff Z-pinch experiments at CESZAR.

I. INTRODUCTION

Z pinches have been studied for decades[1] and involve a cylindrical plasma imploding onto its symmetry axis via a Lorentz force that originates from a current pulse. Several variants of this concept have been realized through the years by modifying the properties of the target plasma and adding liner materials to assist and mediate the implosion. A particular variant of such systems is the gas-puff Z-pinch (GPZP) concept, in which a cylindrical gas flux is launched, with the help of gas nozzles, to surround the target material,[2] which is typically also a gas flux from a separate nozzle. Gas-puff Z pinches (GPZP) have a broad range of applications in high-energy-density physics as a source of intense x rays and neutrons, as well as fusion energy.[1,2] Modern pulsed-power drivers can achieve plasma temperatures of hundreds to thousands of electron volts, while compressing the target gas to solid densities. As a result, the extreme plasmas of GPZP are of interest both to the broader fusion and laboratory astrophysics communities as useful scientific platforms for fundamental physics studies.[3]

One such pulsed-power driver is the Compact Experimental System for Z pinch and Ablation Research (CESZAR) at the University of California, San Diego (UCSD).[4,5] Recent experiments at CESZAR have driven GPZP with a linear transformer driver (LTD) that achieves 0.5 MA of current with less than 200-ns rise times without pulse compression, demonstrating high energy-coupling efficiency to the load. These properties, combined with the high repetition rates that can be realized with LTD drivers, make CESZAR an ideal platform for scanning the parameter space of GPZP implosions and for benchmarking simulation codes.

While the ability to use simulation tools to design and interpret Z-pinch experiments is critically important for the success of said experiments, the physics and algorithmic capabilities needed to model such complex systems are often out of reach of open simulation codes. Further, the experimental validation of the needed code capabilities is an arduous and lengthy process that few development teams undertake. There exist simulation codes in the pulsed-power community that have demonstrated their ability to simulate Z pinches, such as MACH2,[6] PERSEUS,[7] FLUXO,[8] GORGON,[9] and HYDRA.[10] For high-fidelity, validated codes whose usage is more regulated, the sharing approach of a tool such as HYDRA provides us with an effective paradigm: academic teams and students can obtain permission to use HYDRA in their modeling efforts and design their experiments with increased confidence. The verification and validation of publicly available, general-purpose simulation codes like FLASH, on the other hand, has the potential of broadening academic engagement with HEDP and ICF modeling, given their large user-bases.

Here, we present the new Z-pinch simulation capabilities of the publicly available, multi-physics, high-performance-computing code FLASH[11,12] applied to the GPZP concept. We detail the new pulsed power driver and present the first 2D FLASH simulations of GPZPs. The FLASH code has a decade-long track record in radiation hydrodynamic and magneto-hydrodynamic (MHD) modeling of laser-driven, high energy density physics (HEDP) experiments,[12] and was recently expanded to model pulsed-power experiments. We focus primarily on one of the key features of Z-pinches configurations: the onset and development of the magneto-Rayleigh–Taylor instability (MRTI)[13] at the liner surface during the liner compression and subsequent implosion. Being able to accurately capture the growth rate and characteristic scales of this instability will act as a critical assessment of the reliability of FLASH simulations. This study constitutes a first verification of FLASH Z-pinch modeling capabilities in the context of GPZP by executing a detailed code-to-code comparison with previously published results from HYDRA simulations of single- and double-liner GPZP experiments.[14] In that study, the authors demonstrated how the addition of a second liner enhanced the stabilization of their Z-pinch implosions, without significantly reducing the expected yield.

The paper is organized as follows: First, we introduce the radiation-MHD capabilities of the FLASH code and discuss the problem setup. Then we detail the current-drive and circuit-model units for GPZP modeling, as well as the pseudo-vacuum treatment we employ, a constraint inherent to Eulerian codes. Next, we present one- and two-dimensional simulations of single- and double-liner implosions with FLASH and HYDRA, and compare simulated plasma properties, MRTI growth rates, and MRTI characteristic scales against previously published results. We conclude with a discussion on the code-to-code comparison, its limitations, and with an evaluation of FLASH's ability to model GPZP configurations.

II. FLASH CODE

FLASH[11] is a publicly available,[15] parallel, multi-physics, adaptive mesh refinement (AMR), finite-volume Eulerian hydrodynamics and MHD simulation code, developed at the University of Rochester by the Flash Center for Computational Science. Over the past decade and under the auspices of the U.S. DOE NNSA, the Flash Center has added in FLASH extensive HEDP and extended-MHD capabilities[12] that make it an ideal tool for the multi-physics modeling of Z-pinch platforms. These include multiple state-of-the art hydrodynamic and MHD shock-capturing solvers,[16] three-temperature extensions[12] with anisotropic thermal conduction that utilizes high-fidelity magnetized heat transport coefficients,[17] heat exchange, flux-limited multi-group radiation diffusion, tabulated multi-material equation of state (EOS) and opacity, laser energy deposition, circuit models,[18] and numerous synthetic diagnostics.[19] FLASH's newest algorithmic developments include a complete generalized Ohm's law that incorporates all extended MHD terms of the Braginskii formulation.[20] The new extended MHD capabilities are integrated with state-of-the-art resistivity and thermo-electric transport coefficients,[21] developed with support from the U.S. DOE ARPA-E BETHE program.

The FLASH code and its capabilities have been validated through benchmarks or direct application to numerous laser-driven plasma physics experiments[22–28] and have also been verified through code-to-code comparisons.[29–31] For pulsed-power experiments, FLASH has been able to reproduce past analytical models,[32] has been applied in the modeling of capillary discharge plasmas,[33] has been compared to MACH2 simulations of the staged Z-pinch concept,[34] and is currently being validated against gas-puff experiments at UCSD's CESZAR[4,5] and Cornell's COBRA[35] pulsers.

For the simulations presented here, we use the single-fluid, three-temperature (3T) extended MHD ansatz of FLASH,[12] considering three different materials for the gas-puff target, the liner, and for the surrounding "vacuum." The latter is necessary in Eulerian simulations to mediate the transfer of the magnetic field from the outer domain boundary, the values of which are specified through a current drive circuit model, discussed below. The equations of state and opacity for each material use tabulated PROPACEOS[36] data. The 3 T MHD ansatz is coupled with heat-exchange between ions and electrons as well as FLASH's operator-split, multi-group, flux-limited radiation diffusion.[12] The FLASH code solves separate ion, electron, and radiation internal energy equations with a finite volume discretization that conserves the total energy (for more information see Tzeferacos *et al.*[12]) We use six radiation-energy bins spanning 10^{-1}–10^5 eV. The calculations include anisotropic thermal conductivity and magnetic resistivity (Spitzer model). Both are solved implicitly, leveraging HYPRE,[37,38] which greatly relaxes parabolic time step restrictions.

A. Simulation setup

We performed one- and two-dimensional FLASH simulations of double and triple nozzle CESZAR experimental configurations. The results are compared to previously-published 2D HYDRA simulations,[14] with a particular focus on the dynamics of the MRTI. The setup consists of annular single and double Neon liners on a Deuterium target. The initial density profiles are radially Gaussian. The target has a full-width half-maximum of 1 cm and a peak density of 1 μg cm^{-3}. The liners are centered at radii of 1.25 and 2.5 cm, with full-width half-maximum of 0.5 cm, and peak densities of 1 and 0.5 μg cm^{-3}, respectively. For the single liner case, the inner liner is replaced by extending the target with a constant density profile at 0.05 μg cm^{-3}. The axial length of the plasma column is 1 cm. The initial target and liner profiles are presented in Fig. 1. The interface between target and liner is located where the mass fraction of each is at 50%. As a result, the target initial radius is 2.25 cm for the single liner case and 1 cm for the double liner case. Then, the total mass of the single liner

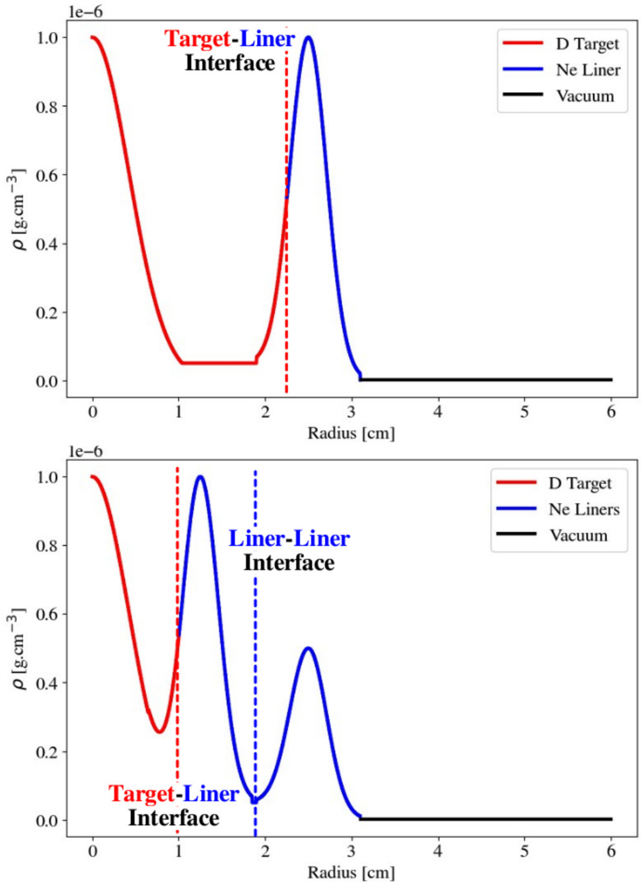

FIG. 1. 1D profiles of the initial mass density for the single liner (top) and the double liner (bottom) configurations. The target material is in red, the liner material in blue, and the "vacuum" material in black. The target–liner interface used to calculate convergence ratios is indicated in a red dashed line, the liner–liner interface in blue dashed line and the liner-vacuum interface is located at 3.1 cm, in both cases.

configuration is 10.11 μg cm^{-3} whereas it is 9.87 μg cm^{-3} for the double liner setup.

The 1D and 2D FLASH simulations have cylindrical domains with a radial extent of 6 cm and are solved onto uniform grids. At the outer radial boundary, the circuit boundary condition is applied. The latter consists of an outflow condition for hydrodynamic variables and the toroidal magnetic field is computed from a current, solved in the circuit model. The hydrodynamic solver uses second order piece-wise linear reconstruction, with the MC (monotonized central) slope limiter, and the HLLD (Harten-Lax-van Leer-Discontinuities[39]) Riemann solver. The 1D simulations use 1280 grid points over the radial domain and a CFL of 0.8. The 2D simulations use square grid cells, with 480 grid points radially and 80 grid points axially, for an axial domain length of 1 cm. The Courant–Friedrichs–Lewy (CFL) number used is 0.4. While the HYDRA simulations in Narkis *et al.*[14] also studied cases where an axial magnetic field was initialized, at varying magnitudes, such cases were not considered here. This work constitutes the first verification of FLASH capabilities to simulate Z-pinches physics and accurately capture MRTI dynamics.

B. Current drive

The current drive unit in FLASH implements the external current-generated magnetic field in Z-pinch experiments. For a given time-dependent current profile, the toroidal magnetic field is calculated according to Ampère's law and is enforced as a boundary condition at the edge of the simulation domain. The current profile can either be explicitly specified from a file or modeled self-consistently in a circuit, for which the simulation domain is a single circuit element with a time-varying potential. The potential is calculated from Faraday's law and is proportional to the temporal rate of change in magnetic flux over the simulation domain. The CESZAR circuit model[4] was implemented in FLASH for the scope of this work. The CESZAR linear transformer driver (LTD) cavity enclosure consists of 20 parallel capacitor-switch "bricks," with an equivalent drive capacitance of $C_d = 580$ nF, a resistance of $R_d = 15$ mΩ, and an inductance of $L_d = 8.5$ nH. A shunt resistance of $R = 1.9\,\Omega$ models the current loss around the cavity body. The load loop of the circuit model consists of the vacuum feed resistance and inductance, and the load voltage is calculated across the simulation domain. Experimental results,[4] based on a 25-mm-radius and 20-mm-long aluminum cylinder target with a 6-cm return radius, suggest a total pulse inductance of 22 nH and resistance of 21 mΩ. After accounting for a short-circuit target inductance of 3.5 nH, the equivalent load loop circuit inductance is $L_l = 10$ nH and resistance is $R_l = 6$ mΩ. The circuit capacitor is initially charged to a voltage of $V_0 = 200$ kV, which discharges self-consistently with the target implosion. The circuit model implementation was verified using a separately developed Python solver for the two-loop configuration, where the simulation domain was replaced with the short-circuit vacuum inductance. The implementation was verified using a separately developed Python solver that omits and short-circuits the simulated target.

Figure 2 shows the current profiles for single and double liner Z-pinch 2D FLASH simulations, compared to current profiles from HYDRA.[14] The short-circuit model acts as a current upper bound

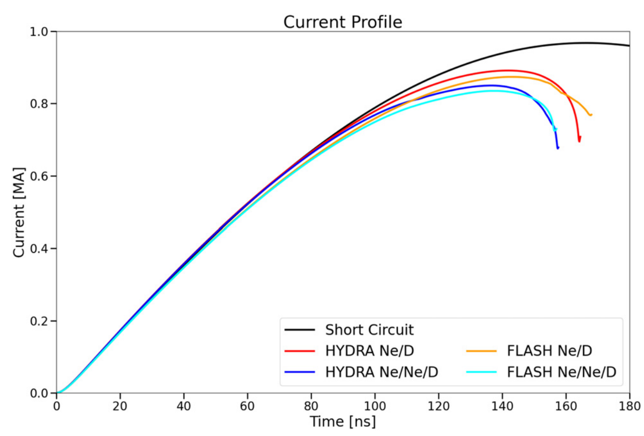

FIG. 2. Current profile comparison between HYDRA and FLASH simulations for single and double liner experiments, showing agreement in rise time and peak current. The cases shown include the short circuit configuration—computed with a Python implementation of the CESZAR circuit (black line), the HYDRA single liner case (Ne/D, red line), the HYDRA double liner case (Ne/Ne/D, blue line), the FLASH single liner case (Ne/D, orange line), and the FLASH double liner case (Ne/Ne/D, teal line).

since it does not include the reactance of the target. The FLASH and HYDRA current profiles are in good agreement, giving approximate rise time of 160 ns and a peak current of 900 kA.

III. SIMULATION RESULTS

A. One-dimensional GPZP simulations

We begin by presenting 1D (radial) simulations of the CESZAR experiment with FLASH. These simulations can be directly compared to and used to verify 2D runs with axially uniform density, wherein the MRTI is not seeded and does not develop. Furthermore, the short time-to-solution of 1D simulations makes them useful for parameter scans and for assessing the new implementations in FLASH, such as the circuit model. We can also note that, since MRTI is not captured in 1D, these results give ideal-compression upper bounds for plasma density and temperature.

The radial length of the domain is 6 cm, and the grid is uniformly-spaced with 1280 grid points. We note that the 1 cm axial length of the plasma column enters in the calculation of magnetic flux for the circuit model. In Figs. 3 and 4, we present the density, radial velocity, toroidal magnetic field, and ion temperature for the single and double liner configurations, respectively. The lineouts are shown at different times to monitor the pinch evolution until stagnation and match the times shown in the respective Figs. 3 and 5 in Narkis *et al.*[14]

Overall, we obtain a pair of conventional gas-puff Z-pinch implosions, characterized by a main shock that is driven by a magnetic piston from the rightmost boundary and collapses onto the axis of symmetry at $R = 0$. The double liner case implodes faster than the single liner configuration, which is consistent with the differences in the initial conditions. By replacing the low-density extended Deuterium target with the Neon inner liner, the overall compressibility of the configuration increases, whereas the main shock will enter the target material later. This also increases the shock strength, as we can see by comparing Figs. 3 and 4. As a result, the double liner case has a higher magnetic field approximately 2× the value obtained in the single liner case, as seen in the bottom right panels of Figs. 3 and 4. This ultimately

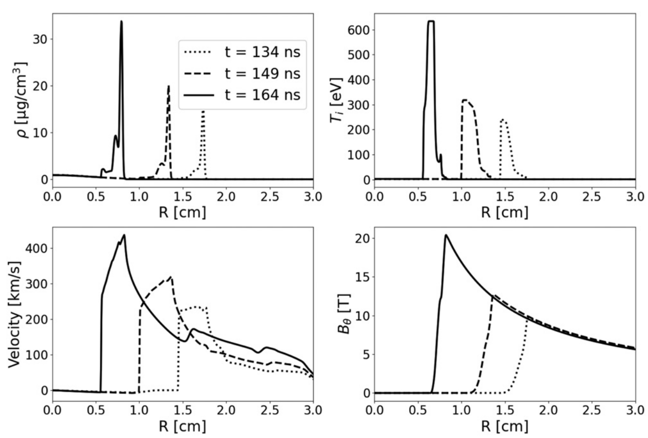

FIG. 3. 1D FLASH simulation at different times in the single liner case. Shown are the plasma density in gram per cubic centimeter (top left), ion temperature in electron volts (top right), radial velocity in kilometer per second (bottom left), and toroidal magnetic field in Tesla (bottom right). The times are chosen to match Fig. 3 in Narkis *et al.*,[14] namely, 134 ns (dotted line), 149 ns (dashed line), and 164 ns (solid line).

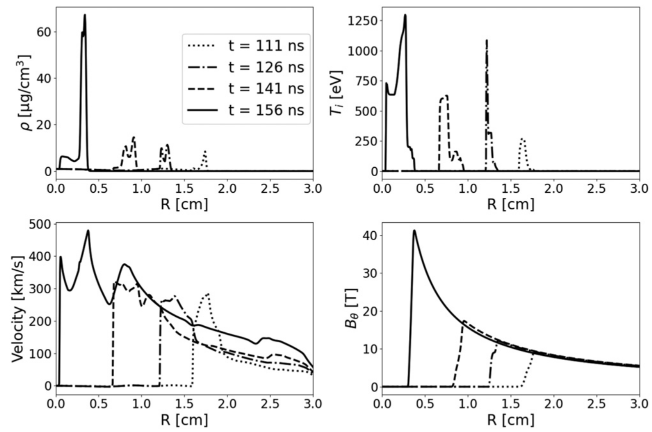

FIG. 4. Same as Fig. 3 but for the double liner configuration. The times are chosen to match Fig. 5 in Narkis *et al.*,[14] namely, 111 ns (dotted line), 126 ns (dot-dashed line), 141 ns (dashed line), and 156 ns (solid line).

results in higher densities and temperatures in the Deuterium target, with the latter reaching 1.25 keV at t = 156 ns (Fig. 4, top right). We later compare the characteristic quantities obtained at peak compression against HYDRA data, for a first assessment of FLASH's Z-pinch capabilities in 1D.

While little can be inferred on how these setups behave in terms of the MRTI, which will be triggered in 2D, we can inspect how the mass averaged ion temperature evolves with respects to convergence ratio (CR) of the target—a common way of looking deeper into the dynamics of Z-pinch compression. Particularly high convergence ratios, here defined as the ratio of initial liner inner radius over its radius at peak compression, are expected to lead to particularly unstable configurations in 2D. These curves are shown on Fig. 5.

From this plot, we see that the single liner configuration compresses the target 2–3× more than the double liner configuration. The addition of the inner liner between the target and the outer liner prevents the implosion from reaching high CR values (∼ 70 in single liner peak compression vs 25 in double liner peak compression). The evolution of the mass averaged ion temperature inside the target also shows a pair of jumps at CR $\simeq$ 3 and $\simeq$ 8 for the double liner case and a single jump at CR $\simeq$ 10 for the single liner case. These jumps are thought to be due to nonadiabatic effects of various kinds. While certain studies have discussed the influence of preheating by radiative waves,[40] radiation effects are not sufficient in these configurations to affect a change in the adiabat.

Instead, these jumps can be economically attributed to the initial conditions of the liner-target and liner-liner interfaces and, in particular, pressure discontinuities that can result in entropy-increasing compression waves. The single liner configuration utilizes the same Gaussian profile for the target as in the double liner case, but a constant fill of $1.5 \times 10^{16}\ \mathrm{cm}^{-3}$ of deuterium is used instead of the innermost liner. The liner-target interface in both cases results in the first compression wave observed in the 1D density profiles, that, upon reaching the pinch axis, heats part of the target plasma at a nearly constant CR. In the double liner case, part of the target material mixes and expands up to the liner-liner interface. The different mass distributions and material properties in the shock paths therefore will result in

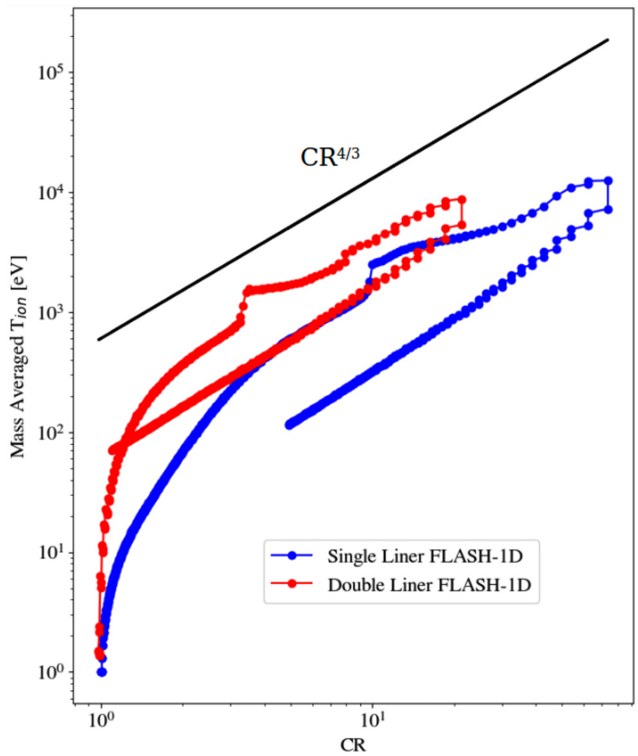

FIG. 5. Mass averaged ion temperature evolution in the target as a function of convergence ratio (CR) in the 1D FLASH simulations. The solid black curve shows the theoretical adiabatic evolution of the ion temperature, $T_i \propto \mathrm{CR}^{4/3}$.

modified shock dynamics between the single and double liner setups. In particular, we see a secondary compression wave, weaker than in the single liner case since the mass of the target material at the liner-liner interface can be neglected, explaining the secondary weak temperature jump at CR $\simeq 8$. Ultimately, the targets follow a similar cooling trend, which can be explained by the similarities in thermodynamical properties of the targets in both setups at peak compression as described in Tables I and II. The apparent discrepancies in CRs are due to the choice of initial target radius as described in the initial conditions section. In particular, if we were to consider the same initial target radius for the single-liner case as that of the double-liner case (i.e., 1 cm), the corresponding CR would be ~ 31, much closer to the CR of ~ 25 we get in the double-liner case. Overall, the addition of a secondary inner liner does not modify greatly the target properties, but is expected to significantly reduce MRTI at the outer liner-vacuum interface.

TABLE I. Target quantities at peak compression in the single-liner simulations, comparing 1D and 2D FLASH and HYDRA simulations.

	1D FLASH	2D FLASH	1D HYDRA	2D HYDRA
t (ns)	180.8	164.4	176.3	164.2
r (μm)	258	1138	302	1954
n_i (cm^{-3})	6.69×10^{19}	3.60×10^{19}	6.26×10^{19}	1.02×10^{19}
T_i (keV)	7.73	10.65	8.18	6.35
T_e (keV)	0.97	0.531	2.81	3.48

TABLE II. Target quantities at peak compression in the double-liner simulations, comparing 1D and 2D FLASH and HYDRA simulations.

	1D FLASH	2D FLASH	1D HYDRA	2D HYDRA
t (ns)	162.6	156.9	159.3	156.5
r (μm)	305	824	794	832
n_i (cm^{-3})	6.01×10^{19}	1.06×10^{19}	9.50×10^{18}	9.01×10^{18}
T_i (keV)	7.55	3.20	7.39	7.62
T_e (keV)	0.94	0.43	1.33	1.66

B. Two-dimensional GPZP simulations

In order to verify this, two-dimensional cylindrical simulations were performed of the double and triple nozzle CESZAR experiment (i.e., single and double Ne liners). The simulation domain extends 6 cm radially and 1 cm axially on a uniform grid of square cells (viz., aspect ratio of 1). A resolution choice of $[480 \times 80]$ for the FLASH simulations is made to match the axial resolution used in Narkis *et al.*[14] for their HYDRA simulations, which resolves the dominant mode of the MRTI that work recovers.

1. Qualitative comparison

Given the relatively low axial resolution, FLASH simulations with unperturbed liner density initialized as a radially smooth Gaussian profile furnish implosions that are in exact agreement with the 1D runs discussed above. To seed the MRTI, the liner is therefore initialized with a random density perturbation at each grid point. Comparisons were made with perturbation amplitudes of 0.1%, 0.5%, 1.0%, 5.0%, and 10%. The best agreement with the HYDRA simulations was seen using an absolute (over or under) perturbation amplitude of 0.5% of the initial density, which is half of what was used in the HYDRA simulations. Figures 6 and 7 show the results of the 2D single- and double-liner simulations, respectively. The density profiles are plotted at different times, chosen to match the previously published HYDRA calculations.[14]

We find that the FLASH simulation results compare well with the HYDRA results in terms of evolution timescales and instability wavelengths while still showing some differences in certain features. First, for the single-liner configuration, we see that the overall dynamics are well retrieved by the FLASH simulation. The size of MRTI spikes as well as their temporal evolution are also well recovered. One more similarity worth noting is the characteristic size of the large bubble appearing in the nonlinear regime of the instability at around t = 164 ns. This bubble is approximately the same size and has a similar structure in both simulations. However, we also see that, in the FLASH result, the bubble has not yet reached the symmetry axis whilst in the HYDRA simulation it has, thus disrupting the pinch. Overall, sharper features are observed in the HYDRA simulations, which can be partially attributed to the Lagrangian nature of the code that better captures interfaces, especially in such low resolutions.

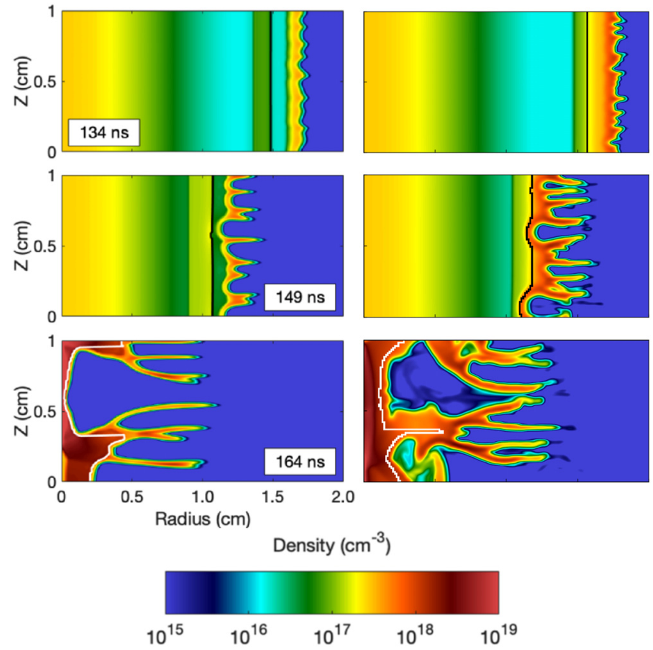

FIG. 6. Ion number density plots from HYDRA (left panels) and FLASH (right panels) for the single-liner implosion simulations at selected times (134, 149, and 164 ns), which span the liner compression dynamics and match Fig. 3 in Narkis *et al.*[14] The same color scale is used for all plots. The interfaces between liners and between target and inner liner are respectively denoted by solid black or white lines. The left panels are reproduced with permission from the authors of J. Narkis *et al.*, Phys. Rev. E **104**, L023201 (2021). Copyright 2011 American Physical Society.

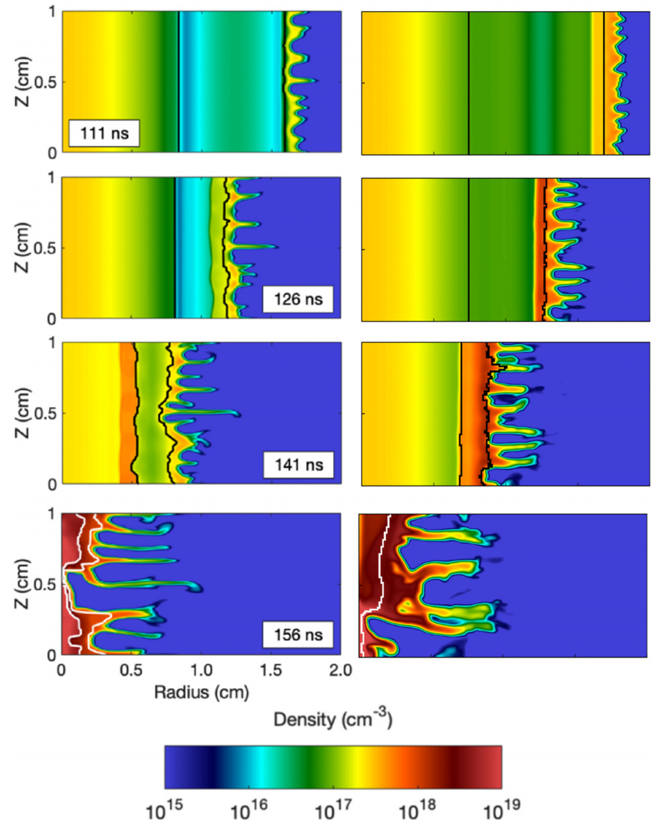

FIG. 7. Same as Fig. 6 for the double-liner implosion simulations at selected times (111, 126, 141 and 156 ns), which span the liner compression dynamics and match Fig. 5 in Narkis *et al.*[14] The same color scale is used for all plots. The interfaces between liners and between target and inner liner are respectively denoted by solid black or white lines. The left panels are reproduced with permission from the authors of J. Narkis *et al.*, Phys. Rev. E 104, L023201 (2021). Copyright 2011 American Physical Society.

The double-liner configuration shows more clear differences in the solutions recovered by the two codes, since there is now one more interface to capture, between the two liners. During the compression, the FLASH simulation does not recover the same density gradient between the liner/target and liner/vacuum interfaces. As more material piles up, the simulations have similar behavior in terms of growth of the MRTI fingers, despite the different density gradients. At $t = 156$ ns, the implosion in the FLASH simulation seems to lag slightly behind, even if the shape of the MRTI fingers is still comparable. A detailed analysis of key simulation results, such as quantities at peak compression and MRTI growth rates, is now presented to assess how the FLASH simulations quantitatively compare to the HYDRA solutions.

2. Quantitative comparison of implosion dynamics

The 2D density distribution can be averaged along Z to recover a 1D radial profile and then compared to equivalent 1D simulations. As previously mentioned, the unperturbed 2D implosions in FLASH evolve exactly as in 1D. When the liners are perturbed and MRTI develops, the pinches no longer recover the 1D peak compression. The position of the peak and the values of density and temperature near stagnation are tabulated for 1D and 2D (axially averaged) HYDRA and FLASH simulations. The results are shown in Tables I and II for the single- and double-liner configurations, respectively.

In the 2D simulations, we defined peak compression as the mean target radius obtained when the liner reaches the symmetry axis. The densities and temperatures are mass-averaged in the target. For both configurations, the implosion dynamics are similar with both codes. The FLASH simulations exhibit a slightly higher target compression, especially for the 1D double-liner case. Consequently, the target mean ion densities follow similar evolution paths, with the FLASH results exhibiting higher values during the implosion. Overall, the FLASH code is able to model the hydrodynamics phase of the Gas-puff Z-pinch implosion consistently with the HYDRA predictions. The main discrepancy between the two sets of simulations is the mass-averaged ion and electron temperatures. In both configurations, the 1D simulations agree fairly well in terms of ion temperatures, but electron temperatures in the FLASH simulations are considerably lower than what is obtained with HYDRA, a trend that also persists in the 2D results. The differences observed between the two sets of simulations can be attributed to the different EOS and thermal conductivity models, as well as the spectral resolution of the flux-limited multi-group radiation diffusion solves. In particular, the PROPACEOS EOS tables used in the FLASH runs assume local thermodynamic equilibrium (LTE) whereas the HYDRA calculations were executed with non-LTE EOS tables, which could result in different ionization in the materials. The radiation transport resolution in FLASH was 6 energy groups, whereas 30 energy groups were used in the HYDRA calculations. Furthermore, we note that the choice of defining the peak compression of the 2D simulations as the time when the vacuum material reaches the symmetry axis creates an additional source of discrepancy/error. Close to pinch time, very small timing differences can lead to drastic changes in the target's thermodynamic properties. Despite the observed differences in electron thermodynamics, the implosion dynamics are not significantly affected, as is the case for the evolution of the MRTI, discussed below.

3. MRTI growth rate

The 2D distribution can also be averaged over the radial coordinate to give an axial profile. This is then spatially Fourier-transformed to investigate the dominant modes of the MRTI. Here, a dominant mode was selected and the rate of growth of the Fourier amplitude was fit to an exponential function over a ~10 ns window in the linear growth regime before stagnation. The fit was performed on the same temporal scale for both HYDRA and FLASH results. Figure 8 for the single-liner configuration and Fig. 9 for the double-liner configuration compare the MRTI growth rate between the FLASH and HYDRA runs.

In both cases, the dominant mode, its Fourier amplitude, and the MRTI linear growth rate are similar, showing that FLASH can capture the dynamics of MRTI during the implosion as accurately as HYDRA. However, we still note some differences in the Fourier amplitude at late times. The low fixed resolutions used with the Eulerian FLASH calculations cannot capture the short-mode dynamics of the MRTI and therefore the Fourier amplitudes and fitted growth rates are slightly lower than for the HYDRA simulations, but sufficiently close for the dynamics to agree very well in the linear regime. The small differences in dominant modes can also be explained by the resolution used for the Fourier analysis, which differs by a factor of 1.25. Both codes recover lower instability amplitudes in the double-liner configuration than for the single-liner simulation, recovering thus the stabilization effect seen in Narkis *et al.*[14]

Finally, we note that the Fourier amplitude values reported here are two orders of magnitude larger than what was reported for the HYDRA results in Narkis *et al.*[14] This comes from the fact that a numerical factor was added during that analysis, scaling the amplitude to microgram per meter, which we chose to re-normalize here for the comparison. The amplitude values now reflect the total mass per unit length of the setup (around 10 μg cm^{-1}), which is consistent with the level of instability observed in the density maps of Figs. 6 and 7. Overall, we conclude that the low-resolution FLASH simulations agree well with previously published HYDRA simulations in terms of MRTI dynamics.

IV. DISCUSSION

Accurate computational modeling of gas-puff Z-pinch experiments can be critically important to design new experimental platforms and to untangle the physics at play when diagnostics only provide path-integrated measurements. Numerical simulations are leveraged to infer underlying physics and provide estimates of key plasma properties needed to calibrate diagnostics. Platform advancements therefore require verified and validated radiation MHD codes to be broadly available to academic groups.

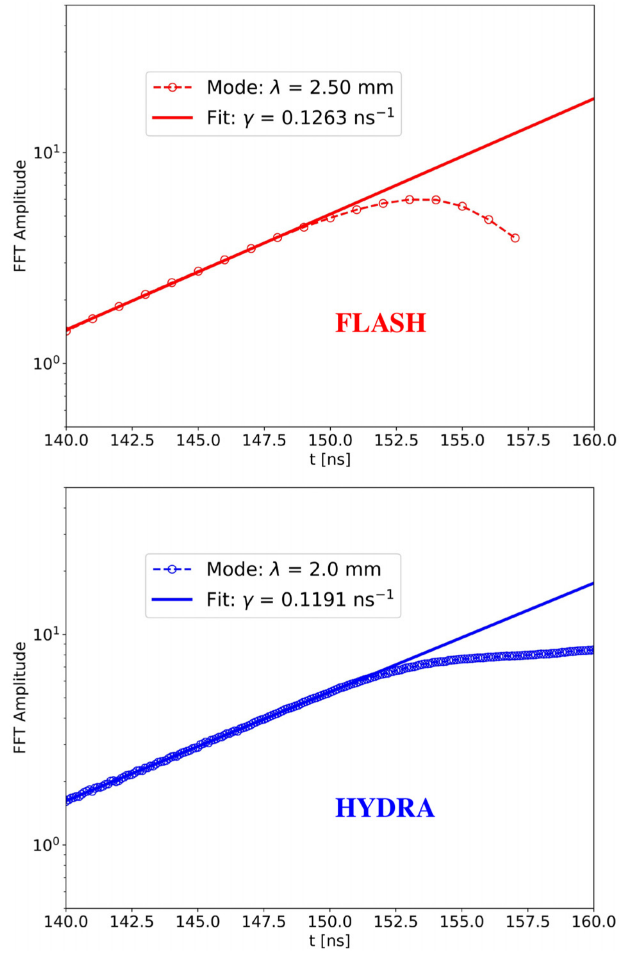

FIG. 8. Fourier amplitude in microgram per centimeter of the dominant mode with an exponential fit, comparing FLASH (top panel) and HYDRA (bottom panel) simulations for the single-liner experiment.

In this work, we presented a verification study of gas-puff Z-pinch FLASH simulations using previously-published results[14] with HYDRA,[10] for single- and double-liner implosion experiments at CESZAR.[4,5] The qualitative agreement seen in this comparison, the similarities of emerging features in both sets of simulations, and the agreement in the growth rate of the dominant MRTI mode in the linear regime build confidence in the new Z-pinch capabilities of FLASH. As noted above, some quantitative differences do remain, which could be attributed to the different EOS and conductivity models used. A much more detailed code-to-code comparison would be required to reconcile these differences, including a study of the single-mode growth in the linear and nonlinear regimes, as well as a study of the multi-mode coupling for different wavelengths and amplitudes.

Another possible improvement would be the use of adaptive mesh, which is available in FLASH, to improve the resolution and the sharpness of the solution close to liner/liner, liner/target, and liner/vacuum interfaces without significantly increasing the time to solution. By using this capability, we can also increase the spatial resolution while keeping the same initial perturbations to enact the comparison with

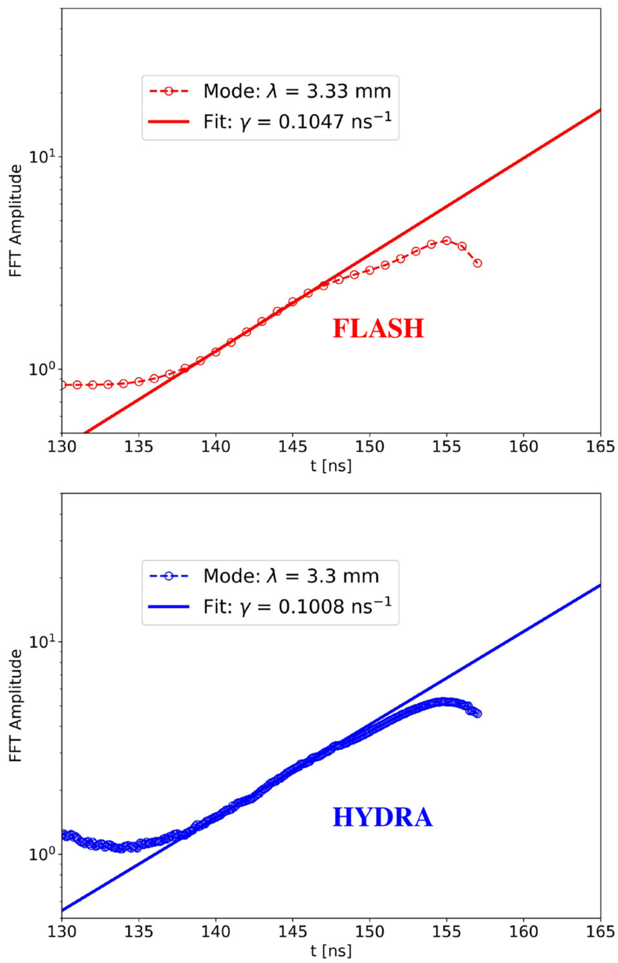

FIG. 9. Fourier amplitude in microgram per centimeter of the dominant mode with an exponential fit comparing FLASH (top panel) and HYDRA (bottom panel) simulations for the double-liner experiment. The amplitude range is set to match the single liner data plot.

the HYDRA results. We are currently conducting a more extensive study on how resolution choices can affect the modeling of gas-puff implosions, which has indicated that the seeding of shorter MRTI wavelengths can affect implosion dynamics. In particular, the growth rate of the dominant mode of the MRTI can drastically increase, leading to highly unstable setups. This will be addressed in a future paper.

Overall, we have been able to recover similar results between the low-resolution FLASH simulations and previously published HYDRA calculations, despite algorithmic differences between the two codes. The next step following this verification is the experimental validation of FLASH against experimental data from CESZAR, on this platform. In doing so, we will show that FLASH constitutes a robust simulation tool for supporting future gas-puff Z-pinch experimental designs and analyses.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

D. Michta: Investigation (equal); Software (equal); Writing – original draft (equal); Writing – review & editing (equal). **P. Tzeferacos:** Funding acquisition (equal); Investigation (equal); Project administration (equal); Software (equal); Supervision (equal); Writing – original draft (equal); Writing – review & editing (equal). **V. Tranchant:** Investigation (equal); Writing – original draft (equal); Writing – review & editing (equal). **J. Narkis:** Investigation (equal); Writing – review & editing (supporting). **E. C. Hansen:** Investigation (supporting); Software (equal); Writing – review & editing (supporting). **M. B. P. Adams:** Software (equal). **K. Moczulski:** Software (equal); Writing – review & editing (supporting). **A. Reyes:** Software (equal); Writing – review & editing (supporting). **F. Conti:** Writing – review & editing (supporting). **F. N. Beg:** Funding acquisition (equal); Project administration (equal); Supervision (equal); Writing – original draft (supporting); Writing – review & editing (supporting).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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